

Face Emotion Detection Using MobileNetV2

Complete Urdu + English Mixed Presentation & Viva Guide

1. Introduction

Yeh project real-time Face Emotion Detection system hai jo MobileNetV2, TensorFlow, OpenCV, Flask aur React.js use karta hai.

2. Problem Statement

Existing systems heavy hain ya real-time nahi hote. Hamara system lightweight aur fast hai.

3. Objectives

Real-time emotion detection, web application aur lightweight deployment.

4. FER2013 Dataset

Dataset mein 35,887 images aur 7 emotions hain: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral.

5. Preprocessing

Grayscale to RGB, Resize 48x48, Normalization aur Train/Test split.

6. Why MobileNetV2?

Sir Question: Why MobileNetV2? Answer: Lightweight, less parameters, fast inference aur transfer learning support.

7. System Architecture

React Frontend -> Flask API -> Haar Cascade -> MobileNetV2 -> JSON Response.

8. API Questions

Sir Question: /predict endpoint kya karta hai? Answer: POST request receive karta hai aur emotion prediction return karta hai.

9. Testing

Unit Testing, Integration Testing, Black Box Testing, Performance Testing.

10. Results

Accuracy 78%, Model Size 15MB, Inference Time 18ms, FPS 15-20.

11. Deep Viva Questions

Q: Why FER2013?

A: Because it is a benchmark dataset.

Q: Why Haar Cascade?

A: Fast face detection for real-time systems.

Q: Why accuracy not 100%?

A: Lighting, side faces, masks and ambiguous expressions affect predictions.

Q: What is Transfer Learning?

A: Using pretrained ImageNet weights and fine-tuning for emotion detection.

Q: What are limitations?

A: Side faces, masks and low lighting reduce accuracy.

12. Presentation Script

Assalam-o-Alaikum Sir. My FYP is Face Emotion Detection Using MobileNetV2. The system detects seven emotions in real-time using Deep Learning and Computer Vision.

13. Extra Viva Questions

Sir: Explain MobileNetV2 architecture.

Sir: Explain preprocessing pipeline.

Sir: Explain Flask and React communication.

Sir: Explain Softmax layer.

Sir: Explain Overfitting and Dropout.

Sir: Explain Confusion Matrix.

Sir: Explain Accuracy, Precision and Recall.

14. Extra Viva Questions

Sir: Explain MobileNetV2 architecture.

Sir: Explain preprocessing pipeline.

Sir: Explain Flask and React communication.

Sir: Explain Softmax layer.

Sir: Explain Overfitting and Dropout.

Sir: Explain Confusion Matrix.

Sir: Explain Accuracy, Precision and Recall.

15. Extra Viva Questions

Sir: Explain MobileNetV2 architecture.

Sir: Explain preprocessing pipeline.

Sir: Explain Flask and React communication.

Sir: Explain Softmax layer.

Sir: Explain Overfitting and Dropout.

Sir: Explain Confusion Matrix.

Sir: Explain Accuracy, Precision and Recall.

16. Extra Viva Questions

Sir: Explain MobileNetV2 architecture.

Sir: Explain preprocessing pipeline.

Sir: Explain Flask and React communication.

Sir: Explain Softmax layer.

Sir: Explain Overfitting and Dropout.

Sir: Explain Confusion Matrix.

Sir: Explain Accuracy, Precision and Recall.